**UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS**

**International General Certificate of Secondary Education**

# MARK SCHEME for the May/June 2011 question paper

# for the guidance of teachers

## 0625 PHYSICS

**0625/21** Paper 2 (Core Theory), maximum raw mark 80

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- Cambridge will not enter into discussions or correspondence in connection with these mark schemes.

Cambridge is publishing the mark schemes for the May/June 2011 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.

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**NOTES ABOUT MARK SCHEME SYMBOLS & OTHER MATTERS**

B marks — are independent marks, which do not depend on any other marks. For a B mark to be scored, the point to which it refers must actually be seen in the candidate's answer.

M marks — are method marks upon which accuracy marks (A marks) later depend. For an M mark to be scored, the point to which it refers **must** be seen in a candidate's answer. If a candidate fails to score a particular M mark, then none of the dependent A marks can be scored.

C marks — are compensatory method marks which can be scored even if the points to which they refer are not written down by the candidate, provided subsequent working gives evidence that they must have known it. e.g. if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows he knew the equation, then the C mark is scored.

A marks — are accuracy or answer marks which either depend on an M mark, or which are one of the ways which allow a C mark to be scored.

c.a.o. — means "correct answer only".

e.c.f. — means "error carried forward". This indicates that if a candidate has made an earlier mistake and has carried his incorrect value forward to subsequent stages of working, he may be given marks indicated by e.c.f. provided his subsequent working is correct, bearing in mind his earlier mistake. This prevents a candidate being penalised more than once for a particular mistake, but **only** applies to marks annotated "e.c.f."

e.e.o.o. — means "each error or omission".

brackets ( ) — around words or units in the mark scheme are intended to indicate wording used to clarify the mark scheme, but the marks do not depend on seeing the words or units in brackets. e.g. 10 (J) means that the mark is scored for 10, regardless of the unit given.

<u>underlining</u> — indicates that this <u>must</u> be seen in the answer offered, or something very similar.

OR/or — indicates alternative answers, any one of which is satisfactory for scoring the marks.

Spelling — Be generous about spelling and use of English. If an answer can be understood to mean what we want, give credit.

Significant figures — Answers are acceptable to any number of significant figures ≥ 2, except if specified otherwise, or if only 1 sig.fig. is appropriate.

Units — Incorrect units are not penalised, except where specified. More commonly, marks are allocated for specific units.

Fractions — These are only acceptable where specified.

Extras — Ignore extras in answers if they are irrelevant; if they contradict an otherwise correct response or are forbidden by mark scheme, use right + wrong = 0

Ignore — Indicates that something which is not correct is disregarded and does not cause a right plus wrong penalty.

Not/NOT — Indicates that an incorrect answer is not to be disregarded, but cancels another otherwise correct alternative offered by the candidate i.e. right plus wrong penalty applies.

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| | | | | |
|---|---|---|---|---|
| **1** | **(a)** | 60.4 – 44.2 | C1 | |
| | | 16.2 ($cm^3$) | A1 | |
| | **(b)** | (density =) mass/volume in any form, letters, words, numbers | C1 | |
| | | 40.5/16.2 e.c.f. | C1 | |
| | | 2.5 e.c.f. | A1 | |
| | | $g / cm^3$ | B1 | |
| | | (accept correct conversion $kg / m^3$, with unit) | | |
| | **(c)** | 60.4 and 40.5 both ticked –1 e.e.o.o. | B2 | [8] |
| **2** | **(a)** | molecules/particles/atoms moving (accept vibrating/oscillating) | C1 | |
| | | molecules colliding (accept with each other) | C1 | |
| | | molecules colliding with walls | A1 | |
| | **(b) (i)** | LH graph – temperature/$T$/$\theta$ / °C/K on horizontal axis<br>RH graph – volume/$V$ / $m^3$/$cm^3$ on horizontal axis | M1 | |
| | **(ii)** | X on LH graph at intersection of line and vertical axis | A1 | [5] |
| **3** | **(a)** | idea that non-renewable sources are finite / get used up | B1 | |
| | **(b) (i)** | solar/sun/sunlight (ignore just light)<br>wind/éolienne accept windmill<br>waves (ignore sea)<br>tidal (ignore sea)<br>hydro(electric) (ignore water)<br>geothermal<br>biomass<br>any 1 | M1 | |
| | **(ii)** | high cost/low effectiveness<br>small output<br>environmental impact<br>cannot be relied upon (wind/solar)<br>any 1 (ignore efficiency) | A1 | |


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| | | | | |
|---|---|---|---|---|
| (c) | (i) | fossil fuel<br>coal<br>oil<br>petrol<br>(natural) gas<br>peat<br>nuclear<br>lignite | any 1 | M1 |
| | (ii) | plentiful/regular/constant/reliable supply<br>cheap/cost effective<br>high output | any 1 | A1 [5] |

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|---|---|---|---|
| 4 | (a) | cool air more dense OR cool air falls<br>OR warm air rises so it can be cooled | B1 |
| | (b) | energy/heat removed from store must be released outside store | B1 |
| | | heat developed by refrigeration unit | B1 |
| | (c) | reduce/prevent heat coming in from outside NOT cold getting out | B1 |
| | | reduce/prevent conduction NOT convection/radiation | B1 |
| | (d) | idea that heat gained from outside = heat removed by refrigeration unit | B2 |
| | | allow B1 for idea of thermostatic control | [7] |
| 5 | (a) | boxes 1 and 4 ticked –1 e.e.o.o. | B2 |
| | (b) | sound/wave reflected/bounces back (from surface) NOT just "returns" | B1 |
| | (c) (i) | cliff A | B1 |
| | (ii) | (s =) vt OR (s =) vt/2 in any form..... allow $s = ut + \frac{1}{2}at^2$ | C1 |
| | | 330 × 1.5 OR 495<br>OR 330 × 0.75 OR 247.5<br>OR 330 × 2.5 OR 825<br>OR 330 × 1.25 OR 412.5<br>OR 330 × 4 OR 1320<br>OR 330 × 2 | C1 |
| | | 660 (m) | A1 |
| | (iii) | both echoes at the same time OR one echo OR louder | B1 |
| | | time value quoted between 1.5s and 2.5s | B1 [9] |

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| **6** | **(a)** | ray bent down at 1st surface, but not beyond/along normal | B1 | |
| | | ray bent down at 2nd surface, but not beyond/along surface | B1 | |
| | | MAX 1 mark if any suggestion of a spectrum shown | | |
| | **(b)** | spot/dot/line AND of one colour accept a single named colour e.g. red | B1 | |
| | **(c)** | spectrum/colours/light dispersed ignore rainbow | C1 | |
| | | red at top <u>and</u> violet at bottom in words in space provided | A1 | [5] |
| **7** | **(a)** | spheres closer together allow touching spheres | B1 | |
| | **(b) (i)** | charging (of anything) by friction/rubbing | B1 | |
| | | plastic/furniture (becomes) charged OR electron/charge transfer | M1 | |
| | | <u>plastic/furniture</u> attracts dust/fluff | A1 | |
| | **(ii)** | idea of charge leaking | B1 | |
| | | water is a conductor | B1 | [6] |
| **8** | **(a) (i)** | parallel | B1 | |
| | **(ii)** | 4.2 (V) | B1 | |
| | **(iii)** | V=IR in any form OR V/R | C1 | |
| | | 4.2 / 3 e.c.f. **(ii)** | C1 | |
| | | 1.4 e.c.f. **(ii)** | A1 | |
| | | A OR amp(s) OR ampere(s) | B1 | |
| | **(iv)** | 1. bigger OR the sum of the two currents OR 2 (A) | B1 | |
| | | 2. same/equal | B1 | |
| | **(b)** | clear series connection of all 3 across battery in one circuit | B1 | |
| | | clear parallel connection of all 3 across battery in other circuit, and must not be shorted out | B1 | |
| | | allow B1 max in (b) if correct series/parallel circuits both shown, but with more or less than 3 resistors in either/both | | [10] |

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| **9** | **(a)** | all 3 lamps in parallel across battery + switch<br>(–1 if any lamps in series, –1 if connections across battery only) | B2 | |
| | **(b)** **(i)** | molecules vibrate over bigger distance OR molecules separate<br>OR bigger space <u>between molecules</u><br>NOT just "molecules need more space"<br>ignore breaking bonds | B1 | |
| | **(ii)** **1.** | bends ignore expands | B1 | |
| | | bends/moves to the right/away from contact/outwards/towards invar strip | B1 | |
| | **2.** | idea that something gets hot | M1 | |
| | | idea that bimetallic strip/invar/brass bends/breaks circuit | A1 | |
| | | idea that something cools (when no current) | M1 | |
| | | idea that bimetallic strip/invar/brass straightens/makes contact | A1 | [9] |
| **10** | **(a)** **(i)** | Fig. 10.1 | B1 | |
| | **(ii)** | Fig. 10.3 | B1 | |
| | **(b)** | 2 complete cycles, any shape (if full-wave rectified, must be 4 humps) | B1 | |
| | | cyclical and equal amplitude above & below axis | B1 | |
| | | uniform spacing | B1 | |
| | | intention of sinusoidal shape accept sinusoidal full-wave rectification | B1 | [6] |
| **11** | **(a)** | thermionic emission | B1 | |
| | **(b)** **(i)** | $S_2$ OR 2 | any 1 correct B1 | |
| | **(ii)** | $S_1$ OR 1 ignore mention of $S_2$ | B2 | |
| | **(iii)** | $S_3$ OR 3 ignore mention of $S_1$ and/or $S_2$ | all 3 correct B2 | |
| | **(c)** | reverse polarity of plates (however expressed)/make upper plate positive<br>OR correct description of use of magnet | B1 | [4] |
| **12** | **(a)** | (radio)activity OR count rate OR counts/s OR particles emitted/s<br>OR rate of decay OR number of <u>undecayed</u> atoms/nuclei<br>OR radiation OR original number of atoms/nuclei | B1 | |
| | | NOT mass/substance/material, unless clearly specified<br>to decrease to half (original value) NOT half the time | B1 | |
| | **(b)** **(i)** | 53 ± 1 (s) | B1 | |
| | **(ii)** | 84 ± 1 (s) | B1 | |
| | **(iii)** | candidate's **(ii)** + candidate's **(i)** | C1 | |
| | | correct evaluation of candidate's **(ii)** + candidate's **(i)** | A1 | [6] |

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